

Q.1₃₇ There are n different gift coupons, each of which can occupy $N(N > n)$ different envelopes, with the same probability $1/N$

P_1 : The probability that there will be one gift coupon in each of n definite envelopes out of N given envelopes

P_2 : The probability that there will be one gift coupon in each of n arbitrary envelopes out of N given envelopes

Consider the following statements

$$(i) P_1 = P_2 \quad (ii) P_1 = \frac{n!}{N^n} \quad (iii) P_2 = \frac{N!}{N^n (N-n)!}$$

$$(iv) P_2 = \frac{n!}{N^n (N-n)!} \quad (v) P_1 = \frac{N!}{N^n}$$

Now, which of the following is true

(A) Only (i) (B*) (ii) and (iii) (C) (ii) and (iv) (D) (iii) and (v)

[Sol. From the given data $n(S) = N^n$; $n(A) = n! \Rightarrow P_1 = \frac{n!}{N^n}$

$P_1 = \frac{n!}{N^n}$ Since the n different gift coupons can be placed in the n definite (Out of N) envelope in ${}^n P_n = n!$ ways

$P_2 = \frac{N!}{(N-n)! N^n}$ As n arbitrary envelopes out of N given envelopes can be chosen in ${}^N C_n$ ways and the n gift coupons can occupy these envelopes in $n!$ ways.]

$$= \frac{N!}{N^n (N-n)!}]$$

Q.2₆₁ The probability that an automobile will be stolen and found within one week is 0.0006. The probability that an automobile will be stolen is 0.0015. The probability that a stolen automobile will be found in one week is

(A) 0.3 (B*) 0.4 (C) 0.5 (D) 0.6

[Hint: $P(S \cap F) = 0.0006$, where S : motor cycle is stolen; F : motor cycle found

$P(S) = 0.0015$

$$P(F/S) = \frac{P(S \cap F)}{P(S)} = \frac{6 \times 10^{-4}}{15 \times 10^{-4}} = \frac{2}{5} \Rightarrow (B)]$$

Q.3₂₅ One bag contains 3 white & 2 black balls, and another contains 2 white & 3 black balls. A ball is drawn from the second bag & placed in the first, then a ball is drawn from the first bag & placed in the second.

When the pair of the operations is repeated, the probability that the first bag will contain 5 white balls is:

(A) $1/25$ (B) $1/125$ (C*) $1/225$ (D) $2/15$

[Hint: 

$$P(E) = P[W B W B] = \frac{2}{5} \cdot \frac{2}{6} \cdot \frac{1}{5} \cdot \frac{1}{6} = \frac{1}{225}]$$

Q.4₇₅ A child throws 2 fair dice. If the numbers showing are unequal, he adds them together to get his final score. On the other hand, if the numbers showing are equal, he throws 2 more dice & adds all 4 numbers showing to get his final score. The probability that his final score is 6 is:

(A) $\frac{145}{1296}$ (B) $\frac{146}{1296}$ (C) $\frac{147}{1296}$ (D*) $\frac{148}{1296}$

[Hint: $P(6) = (51, 15, 24, 42)$ or 11 & $(22$ or 13 or $31)$ or $(22$ & $11)$]

- Q.5₁₁₈ A person draws a card from a pack of 52 cards, replaces it & shuffles the pack. He continues doing this till he draws a spade. The probability that he will fail exactly the first two times is :
 (A) $1/64$ (B*) $9/64$ (C) $36/64$ (D) $60/64$
 [Hint: $P(E) = P(\text{FFS}) = 3/4 \cdot 3/4 \cdot 1/4$]

Q.6 Indicate the correct order sequence in respect of the following :

- I. If the probability that a computer will fail during the first hour of operation is 0.01, then if we turn on 100 computers, exactly one will fail in the first hour of operation.
 II. A man has ten keys only one of which fits the lock. He tries them in a door one by one discarding the one he has tried. The probability that fifth key fits the lock is $1/10$.
 III. Given the events A and B in a sample space. If $P(A) = 1$, then A and B are independent.
 IV. When a fair six sided die is tossed on a table top, the bottom face can not be seen. The probability that the product of the numbers on the five faces that can be seen is divisible by 6 is one.
 (A) FTFT (B*) FTTT (C) TFTF (D) TFFF

[Hint: I. $P(X = 1) = {}^{100}C_1 \left(\frac{1}{100}\right) \left(\frac{99}{100}\right)^{100}$

II. Every key that fits have the same probability = $1/10$

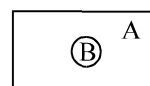
III. Consider $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

but $P(A \cup B) = P(A) = 1$

$1 = 1 + P(B) - P(A \cap B)$

$P(A \cap B) = P(B) = P(B) \cdot P(A) \quad (P(A) = 1)$

IV. Each product 1 2 3 4 5 ; 1 2 3 4 6 ; 1 2 3 5 6 ; 1 2 4 5 6 ; 1 3 4 5 6 ; 2 3 4 5 6 is divisible by six.]



Q.7₁₅₆ An unbiased cubic die marked with 1, 2, 2, 3, 3, 3 is rolled 3 times. The probability of getting a total score of 4 or 6 is

- (A) $\frac{16}{216}$ (B*) $\frac{50}{216}$ (C) $\frac{60}{216}$ (D) none

[Hint: 1, 2, 2, 3, 3, 3 (thrown 3 times)]

$$P(1) = \frac{1}{6} ; P(2) = \frac{2}{6} ; P(3) = \frac{3}{6}$$

$P(S) = P(4 \text{ or } 6) = P(112 \text{ (3 cases) or } 123 \text{ (6 cases) or } 222)$

$$= 3 \cdot \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{2}{6} + 6 \cdot \frac{1}{6} \cdot \frac{2}{6} \cdot \frac{3}{6} + \frac{2}{6} \cdot \frac{2}{6} \cdot \frac{2}{6} = \frac{6+36+8}{216} = \frac{50}{216} = \frac{25}{108}]$$

Q.8₂₀₀ A bag contains 3 R & 3 G balls and a person draws out 3 at random. He then drops 3 blue balls into the bag & again draws out 3 at random. The chance that the 3 later balls being all of different colours is
 (A) 15% (B) 20% (C*) 27% (D) 40%

[Sol. ; $\frac{{}^3C_1 \cdot {}^3C_2}{{}^6C_2} \cdot \frac{{}^2C_1 \cdot {}^1C_1 \cdot {}^3C_1}{{}^6C_3} + \frac{{}^3C_2 \cdot {}^3C_1}{{}^6C_3} \cdot \frac{{}^1C_1 \cdot {}^2C_1 \cdot {}^3C_1}{{}^6C_3}$]

Q.9₁₆₉ A biased coin with probability P, $0 < P < 1$, of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is $2/5$ then the value of P is
 (A) $1/4$ (B) $1/6$ (C*) $1/3$ (D) $1/2$

Q.10₂₀₂ If $a, b \in \mathbb{N}$ then the probability that $a^2 + b^2$ is divisible by 5 is

- (A*) $\frac{9}{25}$ (B) $\frac{7}{18}$ (C) $\frac{11}{36}$ (D) $\frac{17}{81}$

[Hint: Square of a number ends in 0, 1, 4, 5, 6 and 9 favourable ordered pairs of (a^2, b^2) can be (0, 0); (5, 5); (1, 4), (4, 1); (1, 9), (9, 1); (4, 6), (6, 4); (6, 9), (9, 6) and $P(0) = 1/10 = P(5)$; $P(1) = P(4) = P(6) = P(9) = 2/10$]

Q.11₅₂ In an examination, one hundred candidates took paper in Physics and Chemistry. Twenty five candidates failed in Physics only. Twenty candidates failed in chemistry only. Fifteen failed in both Physics and Chemistry. A candidate is selected at random. The probability that he failed either in Physics or in Chemistry but not in both is

- (A*) $\frac{9}{20}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{11}{20}$

Q.12 In a certain game A's skill is to be B's as 3 to 2, find the chance of A winning 3 games at least out of 5.

[Hint: odd in favour of A $3 : 2 \Rightarrow P(A) = \frac{3}{5} = p$; $q = \frac{2}{5}$ [Ans. $\frac{2133}{3125}$]

$$P(E) = {}^5C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^2 + {}^5C_4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right) + \left(\frac{3}{5}\right)^5]$$

Q.13 In each of a set of games it is 2 to 1 in favour of the winner of the previous game. What is the chance that the player who wins the first game shall win three at least of the next four? [Ans. $4/9$]

[Hint: $P(W/W) = \frac{2}{3}$; $P(L/W) = \frac{1}{3}$; $P(W/L) = \frac{1}{3}$; $P(L/L) = \frac{2}{3}$]

Q.14 A coin is tossed n times, what is the chance that the head will present itself an odd number of times?

[Hint: $P(E) = \frac{{}^nC_1 + {}^nC_3 + {}^nC_5 + \dots}{2^n} = \frac{2^{n-1}}{2^n} = \frac{1}{2}$] [Ans. $\frac{1}{2}$]

Q.15 A fair die is tossed repeatedly. A wins if it is 1 or 2 on two consecutive tosses and B wins if it is 3, 4, 5 or 6 on two consecutive tosses. The probability that A wins if the die is tossed indefinitely, is

- (A) $\frac{1}{3}$ (B*) $\frac{5}{21}$ (C) $\frac{1}{4}$ (D) $\frac{2}{5}$

[Sol. Let $P(S) = P(1 \text{ or } 2) = 1/3$
 $P(F) = P(3 \text{ or } 4 \text{ or } 5 \text{ or } 6) = 2/3$
 $P(A \text{ wins}) = P[(S S \text{ or } S F S S \text{ or } S F S F S S \text{ or } \dots) \text{ or } (F S S \text{ or } F S F S S \text{ or } \dots)]$

$$= \frac{\frac{1}{9}}{1 - \frac{2}{9}} + \frac{\frac{2}{27}}{1 - \frac{2}{9}} = \frac{1}{9} \times \frac{9}{7} + \frac{2}{27} \times \frac{9}{7} = \frac{1}{7} + \frac{2}{21} = \frac{3+2}{21} = \frac{5}{21}$$

$$P(A \text{ winning}) = \frac{5}{21}; P(B \text{ winning}) = \frac{16}{21} \text{ Ans.]}$$

Q.16 Counters marked 1, 2, 3 are placed in a bag, and one is withdrawn and replaced. The operation being repeated three times, what is the chance of obtaining a total of 6? [Ans. $7/27$]

[Hint: $P(1 \text{ 2 3 or } 2 \text{ 2 2}) = \frac{6+1}{3^3} = \frac{7}{27}$]

- Q.17 A normal coin is continued tossing unless a head is obtained for the first time. Find the probability that
 (a) number of tosses needed are at most 3.
 (b) number of tosses are even. [Ans. (a) $\frac{7}{8}$, (b) $\frac{1}{3}$]

[Sol. (a) $P(H \text{ or } T H \text{ or } T T H)$; $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$

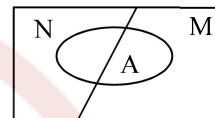
(b) $T H \text{ or } T T T H \text{ or } \dots$; $P(E) = \frac{1/4}{1-1/4} = \frac{1}{4} \times \frac{4}{3} = \frac{1}{3}$]

- Q.18 A purse contains 2 six sided dice. One is a normal fair die, while the other has 2 ones, 2 threes, and 2 fives. A die is picked up and rolled. Because of some secret magnetic attraction of the unfair die, there is 75% chance of picking the unfair die and a 25% chance of picking a fair die. The die is rolled and shows up the face 3. The probability that a fair die was picked up, is

(A*) $\frac{1}{7}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{24}$

[Sol. N = Normal die ; $P(N) = 1/4$
 M = magnetic die ; $P(M) = 3/4$
 A = die shows up 3

$$\begin{aligned} P(A) &= P(A \cap N) + P(A \cap M) \\ &= P(N) P(A/N) + P(M) \cdot P(A/M) \\ &= \frac{1}{4} \cdot \frac{1}{6} + \frac{3}{4} \cdot \frac{2}{6} = \frac{7}{24} \end{aligned}$$



$$P(N/A) = \frac{P(N \cap A)}{P(A)} = \frac{(1/4) \cdot (1/6)}{7/24} = \frac{1}{7} \text{ Ans.]}$$

- Q.19 Before a race the chance of three runners, A, B, C were estimated to be proportional to 5, 3, 2, but during the race A meets with an accident which reduces his chance to $1/3$. What are the respective chance of B and C now? [Ans. $B = 2/5$; $C = 4/15$]

- Q.20 A fair coin is tossed a large number of times. Assuming the tosses are independent which one of the following statement, is True?

- (A) Once the number of flips is large enough, the number of heads will always be exactly half of the total number of tosses. For example, after 10,000 tosses one should have exactly 5,000 heads.
 (B*) The proportion of heads will be about $1/2$ and this proportion will tend to get closer to $1/2$ as the number of tosses increases
 (C) As the number of tosses increases, any long run of heads will be balanced by a corresponding run of tails so that the overall proportion of heads is exactly $1/2$
 (D) All of the above

- Q.21 A and B each throw simultaneously a pair of dice. Find the probability that they obtain the same score.

Hint: [$P[(2 \& 2) \text{ or } (3 \& 3) \text{ or } (4 \& 4) \dots]$] [Ans: $\frac{73}{648}$]

- Q.22 A is one of the 6 horses entered for a race, and is to be ridden by one of two jockeys B or C. It is 2 to 1 that B rides A, in which case all the horses are equally likely to win; if C rides A, his chance is trebled, what are the odds against his winning? [Ans. 13 to 5]

Direction for Q.23 to Q.25

Let S and T are two events defined on a sample space with probabilities

$$P(S) = 0.5, P(T) = 0.69, P(S/T) = 0.5$$

Q.23 Events S and T are:

(A) mutually exclusive

(B*) independent

(C) mutually exclusive and independent

(D) neither mutually exclusive nor independent

Q.24 The value of $P(S \text{ and } T)$

(A*) 0.3450

(B) 0.2500

(C) 0.6900

(D) 0.350

Q.25 The value of $P(S \text{ or } T)$

(A) 0.6900

(B) 1.19

(C*) 0.8450

(D) 0

[Sol. $P(S/T) = \frac{P(S \cap T)}{P(T)} \Rightarrow 0.5 = \frac{P(S \cap T)}{0.69} \Rightarrow P(S \cap T) = 0.5 \times 0.69 = P(S)P(T)$

$\Rightarrow$ S and T are independent **Ans.**

$\therefore P(S \text{ and } T) = P(S) \cdot P(T) = 0.69 \times 0.5 = 0.345$ **Ans.**

$P(S \text{ or } T) = P(S) + P(T) - P(S \cap T) = 0.5 + 0.69 - 0.345 = 0.8450$ **Ans.]**

